

# DNA-MESS: Dynamic Narrative Adjustment via Multi-modal Emotion Sensing & Simulation

Choo Sheriel Shannon\*\*

KAIST

Graduate School of Culture Technology  
Daejeon, Republic of Korea  
shannonchoo@kaist.ac.kr

Murad Eynizada\*\*

KAIST

Graduate School of Culture Technology  
Daejeon, Republic of Korea  
eynizadamurad@kaist.ac.kr

Sang Ho Yoon

KAIST

Graduate School of Culture Technology  
Daejeon, Republic of Korea  
sangho@kaist.ac.kr

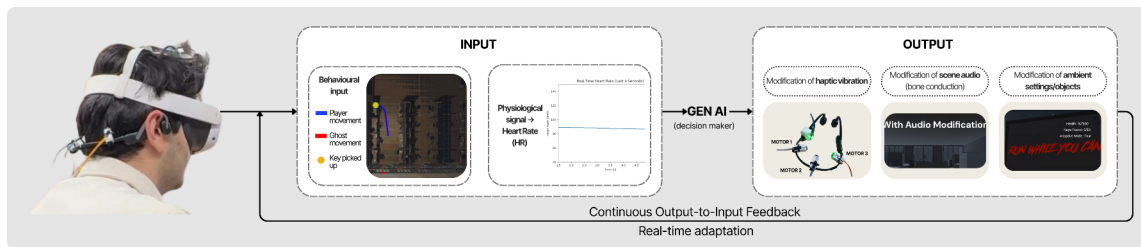


Figure 1: DNA-MESS is a closed-loop adaptive VR system that integrates behavioral and physiological sensing with generative AI to drive real-time multi-modal feedback, including haptics, BC audio, and environmental adaptation

## ABSTRACT

We present DNA-MESS, a closed-loop adaptive VR system that dynamically adjusts narrative and sensory feedback based on users' physiological and behavioral states. A custom hybrid headset integrating PPG sensing, bone-conduction audio, and haptics enables real-time affect-responsive interaction. Signals are processed in Unity and mapped to generative AI-driven adaptations of audio, haptics, and environmental parameters. Unlike existing systems that rely on behavioral inputs or single modalities, DNA-MESS demonstrates a unified multi-modal sensing-feedback pipeline. The system is showcased through an interactive VR horror demo, illustrating the design space of emotionally adaptive virtual environments.

**Index Terms:** Virtual Reality (VR), Affective Haptics, Bone-Conduction Audio, Physiological Sensing, Dynamic Narrative Adjustment, Generative Artificial Intelligence (GenAI)

## 1 INTRODUCTION

Immersive virtual environments aim to evoke presence and emotional engagement, yet many VR experiences remain emotionally static, relying on fixed responses that do not adapt to users' internal states. This limits responsiveness in emotionally charged scenarios such as horror [1, 4].

Prior adaptive VR systems primarily rely on behavioral inputs and often overlook physiological signals. As a result, emotional adaptation is constrained, with audio, haptics, and environmental changes treated as loosely coupled rather than unified feedback channels [2, 3].

To address these gaps, we introduce DNA-MESS, a closed-loop adaptive VR system integrating physiological sensing, behavioral data, and multi-modal feedback. A custom hybrid headset combining photoplethysmography (PPG), bone-conduction audio (BC),

and haptic actuators enables real-time, generative AI-driven adaptation of auditory, haptic, and environmental parameters.

This work focuses on system design and demonstration rather than comprehensive evaluation. The demo showcases DNA-MESS through an interactive VR horror experience, highlighting the design space of emotionally adaptive virtual environments.

## 2 PROPOSED SOLUTION

To enable multi-modal adaptation and physiological feedback, we develop a hybrid headset (Fig. 2) integrating bone-conduction (BC) audio, haptics, and a photoplethysmography (PPG) sensor into a single wearable form. This unified sensing-feedback design serves as the hardware foundation for user-state-driven adaptation powered by generative AI.

Physiological and behavioral signals, including heart data and in-game actions, are processed in real time and passed to a generative AI model that determines adaptive responses without predefined rules. Based on these decisions, the system dynamically adjusts BC audio, environmental sound, and haptic feedback, forming a closed-loop pipeline for emotionally responsive gameplay.

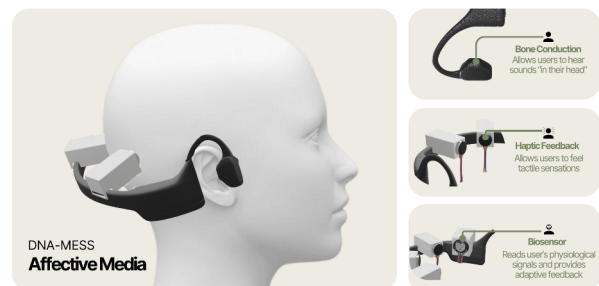


Figure 2: Customized multi-modal headset integrating BC audio, haptic feedback, and physiological sensing.

\*\*These authors contributed equally to this work.

### 3 IMPLEMENTATION DETAIL

This system is implemented as a closed-loop pipeline integrating physiological sensing, generative AI reasoning, and multi-modal feedback within a Unity-based VR environment.

**System Architecture.** The system forms a closed loop linking user-state sensing, generative decision-making, and real-time feedback (Fig. 3). A PPG sensor (115 Hz, Arduino Uno), BC audio output, and three ERM vibration motors collect and deliver feedback. Physiological and behavioral data are processed by a generative AI module that drives adaptive haptic, audio, and environmental responses executed in Unity at runtime.

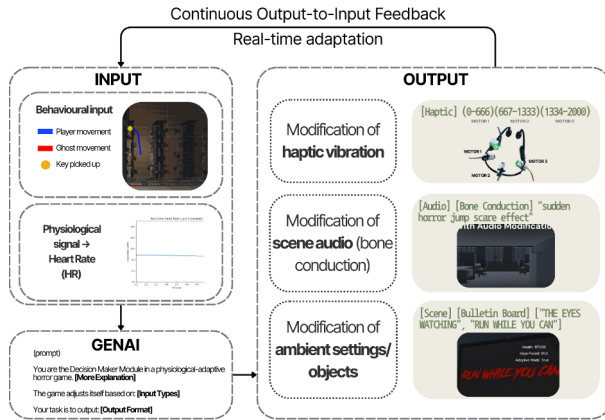


Figure 3: Closed-loop system architecture illustrating how inputs drive real-time adaptation through generative AI.

**Physiological and Behavioral Inputs.** Heart-rate data are computed over a 4-second window and combined with behavioral inputs such as player movement, interactions, and game state. These signals are provided to a generative AI model, which determines adaptive responses without explicit emotion classification and outputs structured control parameters within scene constraints.

**Haptic Feedback.** Haptic feedback is delivered via three ERM vibration motors with fixed amplitude and frequency. Modulation is achieved through timing, duration, and sequencing, enabling distinct tactile patterns that convey proximity and interaction cues.

**Audio and Environmental Adaptation.** The system differentiates internalized BC audio cues from external environmental sounds. The generative AI module also adapts environmental parameters in Unity, including lighting, object activation, and enemy behavior, enabling coherent real-time multi-modal adaptation.

### 4 EXPERIMENT SETUP AND RESULTS ANALYSIS

**Experiment Setup.** A pilot within-subject study was conducted to assess the feasibility of the adaptive system. Five participants experienced both Adaptive and Non-Adaptive conditions in a counterbalanced order. In each condition, participants explored a VR horror scene for 3 minutes while completing a key-finding task, followed by questionnaire-based assessment. The session concluded with brief qualitative feedback.

**Results** Mean iGroup Presence Questionnaire (IPQ) scores showed small directional differences between conditions (Fig. 4). General Presence, Spatial Presence, and Involvement were consistently higher in the Adaptive condition, while Realism was slightly higher in the Non-Adaptive condition. Although no statistically significant effects were observed due to the small sample size, the results suggest a tendency for adaptive multi-modal feedback to enhance perceived presence.

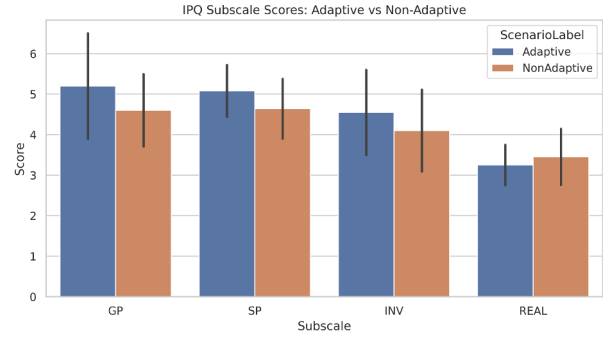


Figure 4: iGroup Presence Questionnaire (IPQ) results comparing Adaptive and Non-Adaptive conditions, including General Presence (GP), Spatial Presence (SP), Involvement (INV), and Realism (REAL).

Positive Affect scores were slightly higher in the Adaptive condition, while Negative Affect scores were comparable across conditions (Fig. 5a). Analysis of fear-related PANAS items showed no difference between conditions (Fig. 5b). Despite limited quantitative separation, all participants reported a preference for the Adaptive condition, indicating perceived experiential benefits not fully captured by questionnaire measures.

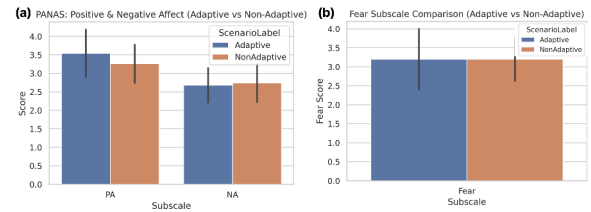


Figure 5: (a) Positive and Negative Affect Schedule (PANAS) scores showing Positive Affect (PA) and Negative Affect (NA). (b) Fear-related PANAS subscale scores (Scared, Nervous, Jittery, Afraid)

### 5 CONCLUSION

This work demonstrates the feasibility of a closed-loop, multi-modal adaptive VR system, with pilot results suggesting increased immersion despite limited quantitative separation. Observed issues related to haptic timing, cybersickness, and hardware constraints highlight the need for refinement. Future work will isolate modality contributions and transition to improved, wireless hardware to support more robust and natural affective adaptation.

### REFERENCES

- [1] S. Bakkes, C. T. Tan, and Y. Pisan. Personalised gaming: a motivation and overview of literature. In *Proceedings of the 8th Australasian Conference on Interactive Entertainment: Playing the System*, pp. 1–10, 2012. 1
- [2] I. Kotsia, S. Zafeiriou, G. Goudelis, I. Patras, and K. Karpouzis. Multi-modal sensing in affective gaming. In *Emotion in Games: Theory and Praxis*, pp. 59–84. Springer, 2016. 1
- [3] J. Marín-Morales, J. L. Higuera-Trujillo, A. Greco, J. Guixeres, C. Llinares, E. P. Scilingo, M. Alcañiz, and G. Valenza. Affective computing in virtual reality: emotion recognition from brain and heartbeat dynamics using wearable sensors. *Scientific reports*, 8(1):13657, 2018. 1
- [4] T. Schubert, F. Friedmann, and H. Regenbrecht. The experience of presence: Factor analytic insights. *Presence: Teleoperators & Virtual Environments*, 10(3):266–281, 2001. 1